

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course:** B.Tech.**Semester :** VII**Branch:** Electrical Engineering / Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. / Electrical & Power Engineering.**Subject Code & Name:** Power System Operation and Control (BTEEC702)**Max Marks:** 60**Date:** 07/02/2025**Duration:** 3 Hr.**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	What is the main cause of voltage instability? a. Generators b. Transformers c. Loads d. Line Losses	L1	1
	Reactive power is measured in ____ a. VA b. VAR c. W d. VAS	L1	1
3	Transmission losses in a line are ____ a. Directly Proportional to Voltage V b. Inversely proportional to Voltage V c. Directly Proportional to V^2 . d. Inversely proportional to V^2 .	L1	1
4	Which type of generator is used in Power System for absorbing Reactive Power? a. Synchronous Generator b. Diesel Generator c. Static Generator d. Brushless Generator	L1	1
	What is the primary objective of Load Frequency Control (LFC) in a power system? a. To regulate Voltage Levels b. To maintain a stable frequency c. To optimize power flow d. To minimize transmission losses	L2	1

6	Which of the following is a characteristic of a brushless excitation system?				L2	1
	a. uses a separate DC generator	b. Uses a rotating rectifier	c. Uses a static power converter	d. Requires a separate exciter generator		
	Which of the following types of excitation systems is most commonly used in modern power plants?				L1	1
	a. Self excited	b. Separately excited	c. Brushless excitation	d. Static Excitation		
8	Which of the following types of stability is concerned with the ability of the power system to return to its original state after a disturbance?				L2	1
	a. Steady state stability	b. Transient stability	c. Dynamic stability	d. Small signal stability		
9	What is the primary objective of economic operation of power system?				L1	1
	a. To minimize transmission losses	b. To maximize generation capacity	c. To minimize operating cost	d. To ensure reliable operation		
10	Which of the following methods is used to improve the voltage Stability of a power system?				L2	1
	a. Increasing the reactive power support	b. Reducing the transmission line impedance	c. Implementing automatic voltage regulators	d. Increasing the generation capacity		
11	What is the purpose of unit commitment in economic operation of a power system?				L2	1
	a. To determine the optimal generation schedule	b. To calculate the transmission losses	c. To evaluate the reliability of the system	d. To determine which generating units to commit to meet the load demand		

12	What is the main objective of optimal power flow (OPF) in economic operation of a power system?				L1	1	
	a. To minimize transmission losses	b. To maximize generation capacity	c. To minimize operating cost	d. To optimize power flow in the transmission network			
Q.2	Solve the following.					12	
A)	Enlist the methods of Voltage Control and Explain any one of them.				L2	6	
B)	Explain the Operation of a Synchronous Condenser with neat diagram.				L3	6	
Q.3	Solve the following.					12	
A)	Classify power system stability, Explain in detail.				L2	6	
B)	Derive swing equation for a synchronous generator.				L3	6	
Q.4	Solve Any Two of the following.					12	
A)	What are the different elements of an excitation system? Explain with the help of block diagram.				L2	6	
B)	Explain the construction and working of DC excitation system with neat sketch.				L3	6	
C)	What are control and protective functions of excitation system? Explain with block diagram.				L2	6	
Q.5	Solve Any Two of the following.					12	
A)	Explain the construction and working of speed governing mechanism with neat diagram.				L3	6	
B)	With a neat block diagram explain the load frequency control of the synchronous machine.				L2	6	
C)	Explain the construction and working of static VAR compensator.				L2	6	

Q. 6	Solve Any Two of the following.		12
A)	Explain the objectives, constraints and solution methods associated with Hydrothermal Scheduling.	L2	6
B)	Derive the condition for economic load dispatch when transmission losses are neglected.	L3	6
C)	Derive the expression for transmission line losses in terms of power plant generation when two units are supplying the load. Also write the equation of loss coefficients.	L3	6
*** End ***			